Sahaj Bhadja

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Summary

Product Software Engineer with 2+ years building **TypeScript/React products**, **Python AI systems**, scalable APIs, and **cloud infrastructure** across authentication, billing, analytics, and real-time workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant services** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured AI evaluation**, model switching, fallback paths, persistent results, and reviewer controls supporting dependable analytical workflows for technical users.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with WebSockets and Azure SQL across distributed locations.
- Reduced resource wastage by 42% through **end-to-end product automation**, implementing JWT access controls, real-time updates, monitoring, automated tests, and Docker/Jenkins deployments across eight modular service domains reliably.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.
- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, C++, Bash
Frontend & Product: React, Next.js, React Native, Angular, HTML, CSS, TailwindCSS, Material UI, UI/UX Design
Backend & APIs: Node.js, Express.js, FastAPI, Flask, Spring Boot, REST APIs, WebSockets, Microservices, JWT, Webhooks
Data & AI: PostgreSQL, MySQL, MongoDB, AWS RDS, Firebase, Redis, Large Language Models, PyTorch, Transformers, Semantic Search, Model Evaluation
Cloud & Infrastructure: AWS, Azure, Google Cloud Platform, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD, Vercel
Engineering: Data Structures, Algorithms, Object-Oriented Programming, System Design, Unit Testing, Integration Testing, Automated Testing, Debugging, Monitoring, Agile, Scrum

Projects

Big-Brain | Live
- Built an **AI-powered collaboration platform** with secure authentication, real-time workflows, semantic search, and ingestion across 50K+ documents, reducing review time by 60% through scalable automated processes reliably.

FinNexus | Live
- Built a **TypeScript fintech platform** supporting 50K+ users and 1M+ transactions with Plaid bank linking, token-based access controls, real-time synchronization, advanced filtering, Kubernetes deployment, and MongoDB-backed data access.

Chat2SQL | Live
- Built an **AI-powered developer tool** across Java and TypeScript, supporting 10+ databases with schema exploration, query generation, Docker, Maven, Node.js, and cross-platform workflows for efficient data access.